

Taewoong Yoon

Department of Physics and Astronomy, Seoul National University
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Education

Seoul National University (SNU), Seoul, Korea

Ph.D Candidate in Physics, Mar 2019 – Present

Thesis Advisor: Prof. Hyunyoung Choi

Pohang University of Science and Technology (POSTECH), Pohang, Korea

Bachelor of Science (BS) in Physics, Mar 2014 – Feb 2019

Seoul Science High School (SSHS), Seoul, Korea

Mar 2011 – Feb 2014

Research Interests

Quantum Sensing, Quantum Simulation, Diamond NV Centers

Research Experience

Korea Institute of Science and Technology (KIST), Seoul, Korea

Visiting Researcher (Advisor: Dr. Junghyun Lee), Feb 2026 – Present

Seoul National University (SNU), Seoul, Korea

Graduate Research Assistant (Advisor: Prof. Hyunyoung Choi), Aug 2019 – Present

Publications

1. **Mesoscopic Spin Coherence in a Disordered Dark Electron Spin Ensemble**, Taewoong Yoon, Sangwon Oh, Junghyun Lee, and Hyunyoung Choi, [arXiv:2602.17074 \(2026\)](#).
2. **Identifying NV center axes via spatially varying microwave fields for vector magnetometry**, Taewoong Yoon, Myungjun Cha, Dohun Kim, and Hyunyoung Choi, *Applied Physics Letters*, **126**, 144002 (2025).

Conferences & Presentations

Oral Presentations

1. “Polarization of P1 electron spin bath via repetitive Hartmann-Hahn transfer,” *APS Global Physics Summit*, Denver, USA, Mar 2026.
2. “Probing and controlling many-body dipolar interactions between electron spins in diamond,” *KPS Spring Meeting*, Daejeon, Korea, Apr 2024.
3. “Precise calibration of magnetic field vector in a diamond nitrogen-vacancy center ensemble,” *Optics and Photonics Congress (OPC)*, Jeju, Korea, Aug 2023.
4. “Aberration correction of femtosecond lasers for deterministic quantum emitters in nitrogen-vacancy diamonds,” *Optics and Photonics Congress (OPC)*, Jeju, Korea, Jul 2021.

Poster Presentations

1. “Deterministic creation of single nitrogen-vacancy center in diamond using femtosecond laser writing,” *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, USA, May 2022.

Honors & Awards

1. BK Excellent TA Award, Jul 2020
2. BK Frontier Fellowship, Mar 2019 – Aug 2019
3. National Scholarship for Science and Engineering, Mar 2014 – Feb 2018